

SAO DEMO CONTROLLER V3.0 to 3.2 – A COMPLETE SAO DEVELOPMENT HOST WITH MULTIPLE POWER OPTIONS!

POWER SOURCES

- (1) USB-C Into RP2040-Zero port (5V)
The default configuration powers/programs the SAO Demo Controller RP2040-Zero regardless of the power switch position. The USB 5V is also routed to the LiPo battery charge IC.
POWER SWITCH ON = downstream SAO is powered.
POWER SWITCH OFF = downstream SAO is not powered.
- (2) External battery, 2-4 AA/AAA or 1S LiPo/LiPoly
(A) The default configuration is for non-rechargeable Primary/Alkaline cells with both the charger and LiPo protector disabled. Allowed input is 3 to 16 V per limits of Buck-Boost 3.3V Converter.
Practically, 3 to 8 "AA" or "AAA" batteries may be used including non-rechargeable Lithium-Iron Disulfide (Li-FeS₂) consumer batteries such as Energizer Ultimate Lithium which may offer up to 1.8V per cell when fresh. A variety of cell cells may be used as long as they can deliver enough current and do not exceed 16V.
(B) Optionally, to use a 1S LiPo/LiPoly cell verify the connection polarity to JST-PH BATT. port. (Adafruit) POWER SWITCH ON = RP2040-Zero and SAO are powered. POWER SWITCH OFF = RP2040-Zero and SAO are not powered.
- (3) SAO Input port 3.3V expected, 2 to 6V allowed
(A) The default configuration expects power to be supplied through the SAO port from the badge or host. As such, solder jumper SJ_REV_PWR_BDG "Reverse Power Badge" is open by default.
(B) As long as the badge/host does NOT have it's own power source, you may optionally provide 3.3V to the badge/host, AKA "reverse powering it" by removing reverse protection diode D4 and closing solder jumper SJ_REV_PWR_BDG. In this configuration, power is provided from your external battery or USB C into the RP2040-Zero through the 3.3V Buck-Boost Converter and into your badge or host device.

Diodes prevent backfeeding from four input power options. Diodes may be removed and solder-bridged to improve efficiency if you make sure you don't create a backfeed power issue between the three sources.

All power sources, with the exception of the USB-C 5V, flow through a switch for convenience. If USB-C power is supplied, the battery charger and RP2040-Zero will always be powered. The switch enables/disables the 3.3V Buck/Boost Converter. SO YOUR SAO WILL NOT GET POWER UNLESS THAT SWITCH IS ON AS WELL!

FOR DEDICATED LIPO ONLY USE

If you want to only use the 1S LiPo and USB power options (likely for Etch SAO Sketch or similar SAOs) remove ALL diodes D1-4.

Solder bridge only the D3 position footpring. These changes will safely provide more runtime and full use of the LiPo capacity.

USB Power will still provide direct power to the RP2040-Zero, and power to the Main Switch via the LiPo charging IC.

LIPO PROTECTOR

LiPo is protected by disconnecting -BATT from ground.

LiPo/Li-Ion CHARGER

For 1S LiPo or Li-Ion cells only.

POWER REV. POLARITY & BACKFEED PROTECTION

0.3A, 0.30V Drop
0.5A, 0.35V Drop
1.0A, 0.45V Drop

MAIN POWER SWITCH

3.3V BUCK-BOOST CONVERTER FOR RP2040-ZERO & SAO STANDARD

IDEAL DIODE

Prevents contention between RP2040-Zero 3V3 LDO output if USB C is connected. If USB C is not connected, 3V3 is supplied to the RP2040 from the external buck-boost converter. Used here to get as close to 3.3V as possible for the RP2040 analog ref.

CURRENT SENSE AMPLIFIER

0.050V drop at 1.00A
*50V/mV = 2.5V output
0.025V drop at 0.50A
*50V/mV = 1.25V output
0.013V drop at 0.25A
*50V/mV = 0.625V output

BADGE GPIO-UART PASS-THROUGH ENABLE

SAO GPIO OR UART PASS-THROUGH SELECT GPIO1 SELECT

I2C HOST PULL-UP

QWIC PORT

QWIC SOCKET
Mfr: JST SM04B-SRSS-TB
Digikey: 455-SM04B-SRSS-TBTR-ND
LCSC: C160404

OUTPUT TO SAO

SAO BADGE/HOST SOCKET
Sullins SFH11-NBPC-D03-ST-BK female header
<https://www.digikey.com/product-detail/en/sullins-connector-solutions/SFH11-NBPC-D03-ST-BK/S9717-ND/4558818>

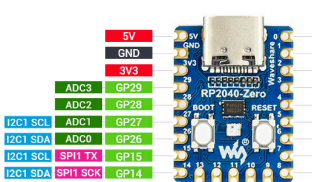
FROM HOST BADGE

SAO CLIENT PLUG
<https://www.digikey.com/en/products/detail/adam-tech/BHR-06-VUA/10414837>

GPIO1 is also data from the badge to the SAO (Badge TX, SAO RX).
GPIO2 is also data from the SAO to the badge (Badge RX, SAO TX).

MICROCONTROLLER: RP2040-ZERO

DEMO CONTROLLER FUNCTIONS:
INPUT 3-5V from Badge or USB-C
GROUND
3V3 @ ~200mA regulated from ZERO
VIN/2 input voltage measurement
SAO output current measurement
SAO GPIO2
SAO GPIO1
USER BUTTON 2
USER BUTTON 1



RP2040-Zero
See <https://www.waveshare.com/wiki/RP2040-Zero> for more information and more GPIO on bottom pads.
Onboard buttons not user accessible.
GPIO16 is NeoPixel onboard.

DEMO CONTROLLER FUNCTIONS:
SAO GPIO2 pass-through or Badge RX / SAO TX
SAO GPIO1 pass-through or Badge TX / SAO RX
SDA1 IN from Badge
SCL1 IN from Badge
SDA0 OUT to SAO
SCL0 OUT to SAO
USER BUTTON 3
USER BUTTON 4
USER 5100 STATUS/UART
GPIO/I2C/UART
USER SPI_CLK/I2C1_SDA/UART
USER SPI_TX_OUT/I2C1_SCL/UART
USER SPI_RX_IN/I2C0_SDA/UART
USER SPI_CS_N/I2C0_SCL/UART

COMMON I2C ADDRESSES

OLED (SSD1306) 128x32 0x3C
OLED (SSD1306) 128x64 0x3D
OLED (SSD1327) 128x128 0x3C

SAO STANDARDS

PRIMARY: <https://docs.google.com/document/d/1EJqyKLMAPsQ9VWF5A4eWol0qMKYr5Giw5rqRmtmM/edit?usp=sharing>
SUPPLEMENT: <https://hackaday.io/project/175182-simple-add-ons-sao>

QWIC/STEMMA RESOURCES

<https://learn.adafruit.com/introducing-adafruit-stemma-qt/what-is-stemma>

REVISION HISTORY

V3.0 (built in July 2026)
* Was not distributed. These were all reworked to at least V3.1.

V3.1 (distributed through August 3, 2026)
* Buck-Boost output voltage setting corrected to 3.3V by changing R6/R7 from 8.2K/2.2K to 4.7K/15K.
* I2C Pull-ups R13&14 reworked from 2.2K to 4.7K for more reliably I2C operation.
* Charging circuit has 2.2K current program resistor, 545 mA. This is a little high for the included LiPo pack. Suggest replacing that resistor with 4.7K for 212mA charge current limit.
* May or may not have V3.1 label added to the board on the top of the USB-C connector.

V3.2 (started distribution August 15, 2026)
* Reduced LiPo charging current, reworked R3 from 2.2K (454 mA) to 4.7K (212mA)
* Add V3.2 label.

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.

Design by Andy Geppert @ www.MachineIdeas.com

Sheet: /
File: SAO_Demo_Controller.kicad_sch

Title: SAO Demo Controller, Micro Badge Development Board

Size: B Date: 2026-08-15

KiCad E.D.A. 10.0.1

Rev: 3.0, 3.1, 3.2

Id: 1/1

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